

Sebastián Cardona

Backend Developer — Python & FastAPI

scarrdona@gmail.com · +57 315 205 2580 · github.com/realProdigium · linkedin.com/in/franchisium · Colombia (Open to Remote)

PROFESSIONAL SUMMARY

Backend developer focused on Python, building production-grade REST APIs, real-time systems, and scalable architectures. Capstone project deployed on a VPS (in 17 days) and Supabase for database with IoT integration, WebSockets, JWT/RBAC authentication, and an ML model. Active Software Engineering student at Universidad Cooperativa de Colombia, graduating 2026.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript (basic), Bash

Frameworks: FastAPI, Django, Flask

Databases: PostgreSQL, SQLAlchemy ORM, Supabase

Auth / Security: JWT, OAuth 2.0, Role-Based Access Control (RBAC)

Real-Time: WebSockets (Starlette), MQTT / aiomqtt, EMQX Broker

DevOps: Docker, Docker Compose, Git, GitHub, Ubuntu / Linux, VPS, CI/CD (basic), Supabase

Testing: Pytest, Postman

Other: Pandas, Power BI, GCP, Pydantic v2, OpenAPI / Swagger

PROJECTS

Real-Time CO₂ IoT Monitoring System | Capstone Project · Production

[FastAPI · PostgreSQL · SQLAlchemy · MQTT/aiomqtt · WebSockets · JWT · RBAC · VPS · HTTPS]

- Designed and implemented a REST API with FastAPI following a layered architecture (routers, services, models, schemas); system deployed on a VPS with HTTPS.
- Integrated an async MQTT client (aiomqtt) with an EMQX broker to receive IoT sensor data from ESP32/Arduino hardware with end-to-end latency < 1 second.
- Built JWT authentication with Role-Based Access Control (RBAC) — admin, operator, and viewer roles — using FastAPI Security dependencies.
- Developed a WebSocket server (Starlette) broadcasting real-time sensor data to all connected clients; delivers the last 50 historical records on connection.
- Modeled a PostgreSQL database (Records, Users) with SQLAlchemy ORM, FK relationships, and connection pooling configuration.

Bookmarks Storage API — MVP | Personal Project

[FastAPI · SQLAlchemy · PostgreSQL · JWT · OAuth 2.0 · X API (Twitter)]

- Built a full CRUD system with JWT authentication and OAuth 2.0 flow with the X (Twitter) API for bookmark import.
- Designed RESTful endpoints with automatic OpenAPI/Swagger documentation and Pydantic v2 data validation.
- Implemented scalable architecture with clear layer separation and database session management.

EDUCATION

B.Sc. Computer Science (Ingeniería de Sistemas)

Universidad Cooperativa de Colombia · In Progress — Expected 2026

LANGUAGES

Spanish: Native

English: Intermediate — Fluent technical reading & writing